

Computer Science BSc; Analysis-2
Questions for the Theoretical Exam (92 questions)
2024/2025 Autumn

1. Define the differentiability and the derivative of a function at a point
2. State the theorem about the connection between the differentiability and the continuity
3. State the derivatives of the following functions: c , $ax + b$, x^n , e^x , $\sin x$
4. State the derivative of the sum
5. State the derivative of the product
6. State the derivative of the quotient
7. State the derivative of the composition (Chain Rule)
8. State the derivative of the inverse (Inverse Rule)
9. State the derivatives of $\tan x$, $\ln x$, a^x
10. Define the arcsin function. State its derivative
11. Define the arctan function. State its derivative
12. Define the concepts of local minimum and of local maximum
13. State the First Derivative Test (First Order Necessary Condition) for local extremum
14. Give an example for a function whose derivative is 0 at a point, however it has no local extreme value at that point.
15. State Rolle's Mean Value Theorem
16. State Cauchy's Mean Value Theorem
17. State Lagrange's Mean Value Theorem
18. State the First Derivative Test for monotonicity
19. State the theorem of L'Hospital's Rule
20. Define the two times differentiability and the second derivative of a function at a point
21. Define the Taylor-polynomial
22. State the connection between the derivatives of a function and of its Taylor-polynomials.

23. State the theorem about the Taylor-formula
24. The Taylor-formula for $n = 0$
25. Define the concepts of strict concavity (strictly concave up, strictly concave down)
26. State the theorem about the characterization of strict concavity using the first derivative. What is the corollary of this theorem using second derivatives?
27. Define the point of inflection
28. Define the antiderivative of a function
29. State the theorem about the set of all antiderivatives of a function
30. State the 5 simple integration rules
31. State the rule of Integration by Parts for antiderivatives
32. State the rule of Substitution Form I. for antiderivatives
33. State the rule of Substitution Form II. for antiderivatives.
34. Define the partition of an interval.
35. Define the lower and the upper sums. Explain the notations used in them
36. Define the lower and the upper integral
37. Define the integrability and the integral of a function
38. Give an example for bounded but non-integrable function
39. Define the oscillation sum
40. State the theorem about the connection between the integrability and the oscillation sums
41. Define the "backward" integration
42. List the properties of the definite integral
43. State the Mean Value Theorem for integral and state its important special case
44. State the theorem about the modification the function at a point
45. State the Fundamental Theorem of Calculus (the Newton-Leibniz Formula)
46. State the rule of Integration by Parts for definite integrals
47. State the rule of Substitution Form I. for definite integral
48. State the rule of Substitution Form II. for definite integral

49. State the theorem about the integrability of the continuous functions
50. Define the Integral Function
51. State the theorem about the continuity of the Integral Function
52. State the theorem about the differentiability of the Integral Function
53. Define the Improper Integral
54. State the Newton-Leibniz formula for improper integrals
55. State the formula for area between two function graphs (with sketch) and the formula for arc length of a function graph (with sketch)
56. Sketch a solid of revolution. State (without proof) the formula for its volume
57. Sketch a solid of revolution. State (without proof) the formula for its surface area
58. Define the space $\mathbb{R}^n$ and the following operations in it: addition, scalar multiple, scalar product
59. Define the norm and the distance in $\mathbb{R}^n$.
60. Define the concept of neighbourhood (environment) in $\mathbb{R}^n$. Sketch the neighbourhoods in $\mathbb{R}^2$.
61. Define: interior point, exterior point, boundary point.
62. Define: the interior of a set, the exterior of a set, the boundary of a set,
63. Define: open set, closed set.
64. When is named a vector sequence to be convergent, and what is the definition of its limit?
65. Define the coordinate sequence. State the theorem about the coordinate-wise convergence of a vector sequence.
66. Define the concept of coordinate-function of a function of type $\mathbb{R}^n \rightarrow \mathbb{R}^m$.
67. Define the accumulation point.
68. Define the limit of a function of type $\mathbb{R}^n \rightarrow \mathbb{R}^m$.
69. Define the continuity of a function of type $\mathbb{R}^n \rightarrow \mathbb{R}^m$ at a point.
70. When is called a function of type $\mathbb{R}^n \rightarrow \mathbb{R}^m$ to be continuous?
71. State the two theorems in connection with continuous functions defined on compact sets in $\mathbb{R}^n$ (compactness of the image, Weierstrass-minimax).
72. Define the differentiability and the derivative of a function type $\mathbb{R}^n \rightarrow \mathbb{R}^m$ at a point.

73. State the theorem about the connection between the differentiability and the continuity for functions of type $\mathbb{R}^n \rightarrow \mathbb{R}^m$
74. Define the (first order) partial derivatives of a function of type $\mathbb{R}^n \rightarrow \mathbb{R}^m$ at a point
75. What are the entries of the derivative matrix for functions of type $\mathbb{R}^n \rightarrow \mathbb{R}^m$ (j -th entry in the i -th row?).
76. Define the concepts of local minimum and of local maximum for functions of type $\mathbb{R}^n \rightarrow \mathbb{R}$
77. State the First Derivative Test (First Order Necessary Condition) of local extremum for functions of type $\mathbb{R}^n \rightarrow \mathbb{R}$
78. Define the two times differentiability and the second derivative (Hesse matrix) of a function type $\mathbb{R}^n \rightarrow \mathbb{R}$ at a point.
79. Define the quadratic form.
80. Classification of quadratic forms: when is called a quadratic form to be positive definite, negative definite, indefinite?
81. How can we classify a two-variable quadratic form in the cases: positive definite, negative definite, indefinite?
82. State the theorem about the second derivative test of local extrema for functions of type $\mathbb{R}^n \rightarrow \mathbb{R}$.
83. Define the intervals in $\mathbb{R}^2$. Give by a picture the partition of the interval $[a, b] \times [c, d]$
84. Define the lower and the upper sums for functions $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$. Explain the notations used in them
85. Define the lower and the upper integral for functions $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$
86. Define the integrability and the integral for functions $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$
87. Write the formula for computation of the double integral for functions $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$ by iterative integrals
88. Define the separable case of the double integral, and give the formula for computation of the integral in separable case
89. Plot an x -normal region in $\mathbb{R}^2$, and give the formula for computation the double integral over x -normal regions
90. Plot a y -normal region in $\mathbb{R}^2$, and give the formula for computation the double integral over y -normal regions
91. Define the polar coordinates in the plane
92. Give the formula of the Polar Transformation of Double Integrals